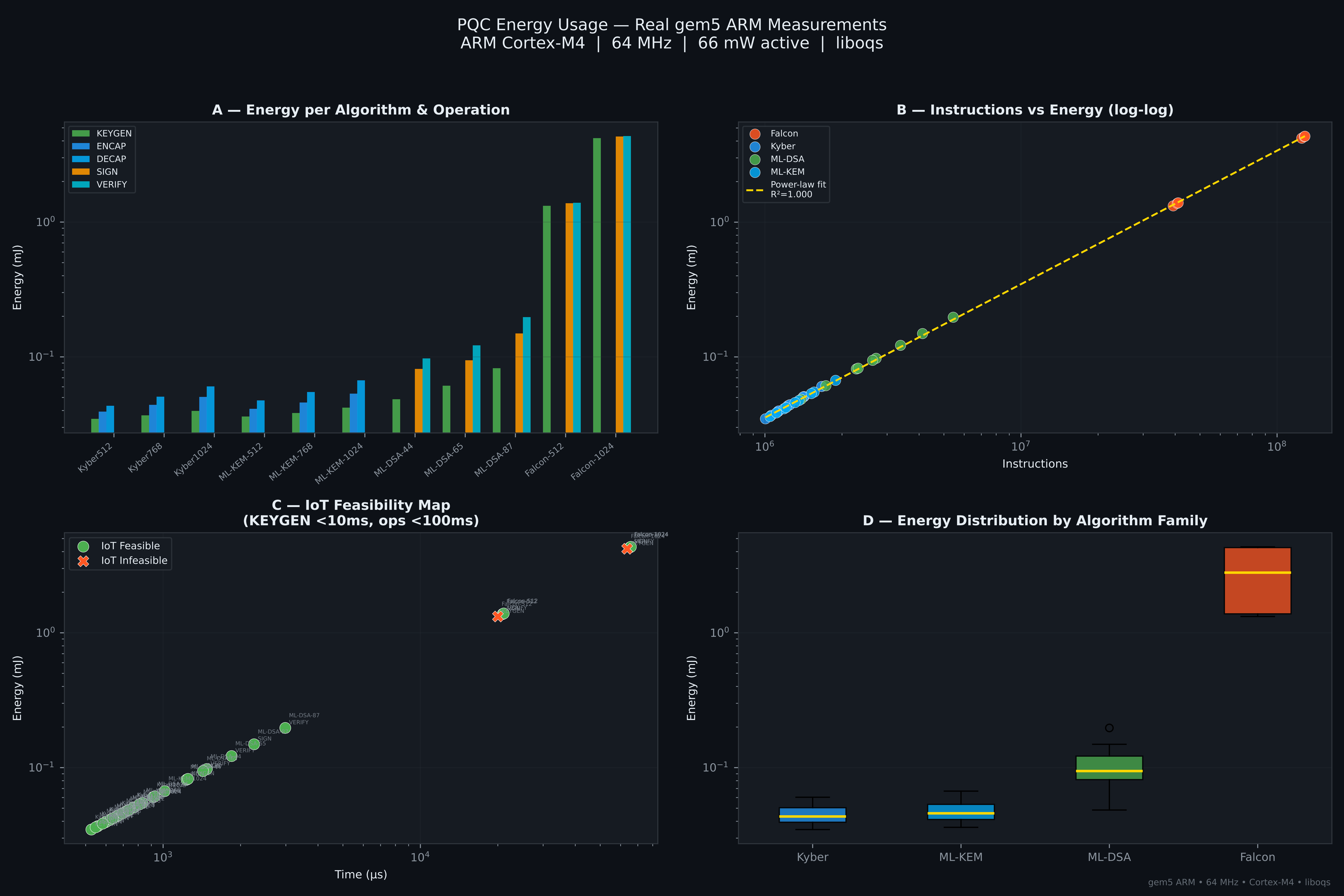
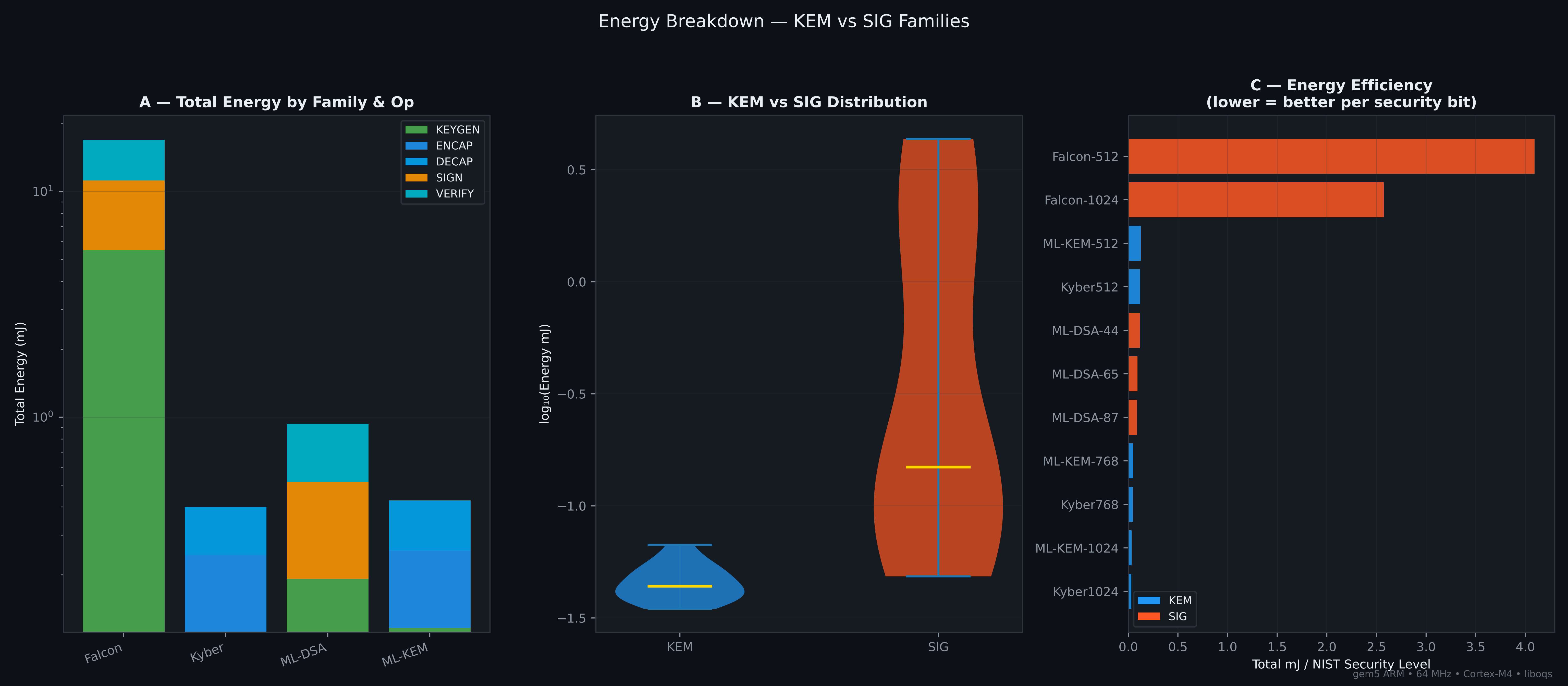
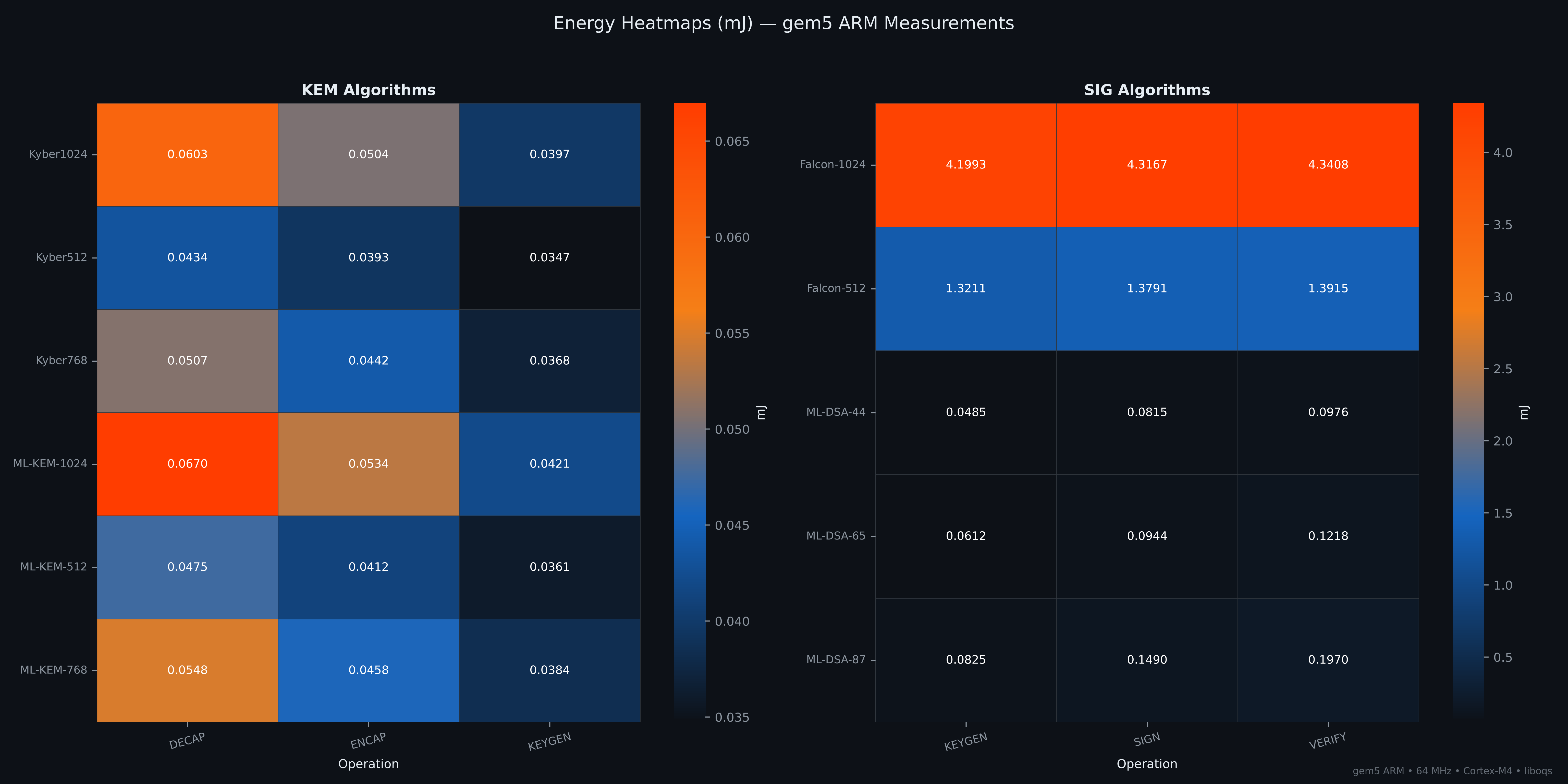
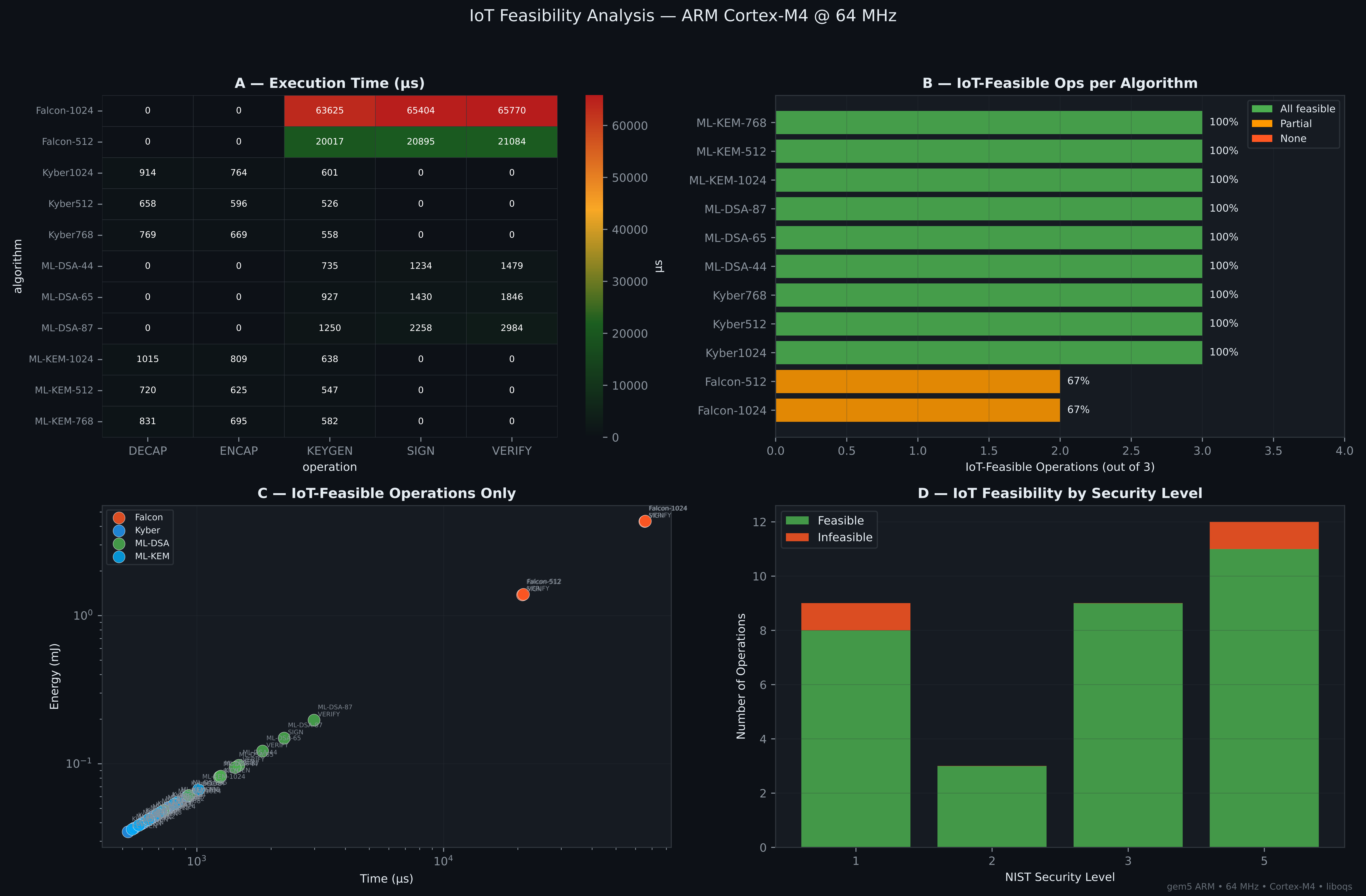

PQC Energy Usage — Real gem5 ARM Measurements
ARM Cortex-M4 | 64 MHz | 66 mW active | liboqs
A — Energy per Algorithm & Operation
KEYGEN
ENCAP
DECAP
SIGN
VERIFY
Energy (mJ)
10^0
10^{-1}
Kyber512
Kyber768
Kyber1024
ML-KEM-512
ML-KEM-768
ML-KEM-1024
ML-DSA-44
ML-DSA-65
ML-DSA-87
Falcon-512
Falcon-1024
B — Instructions vs Energy (log-log)
Falcon
Kyber
ML-DSA
ML-KEM
Power-law fit
R^2=1.000
Energy (mJ)
Instructions
10^6
10^7
10^8
C — IoT Feasibility Map
(KEYGEN <10ms, ops <100ms)
IoT Feasible
IoT Infeasible
Energy (mJ)
Time (μs)
10^3
10^4
Falcon-1024
Falcon-512
ML-DSA-87
VERIFY
D — Energy Distribution by Algorithm Family
Energy (mJ)
Kyber
ML-KEM
ML-DSA
Falcon
gem5 ARM • 64 MHz • Cortex-M4 • liboqs

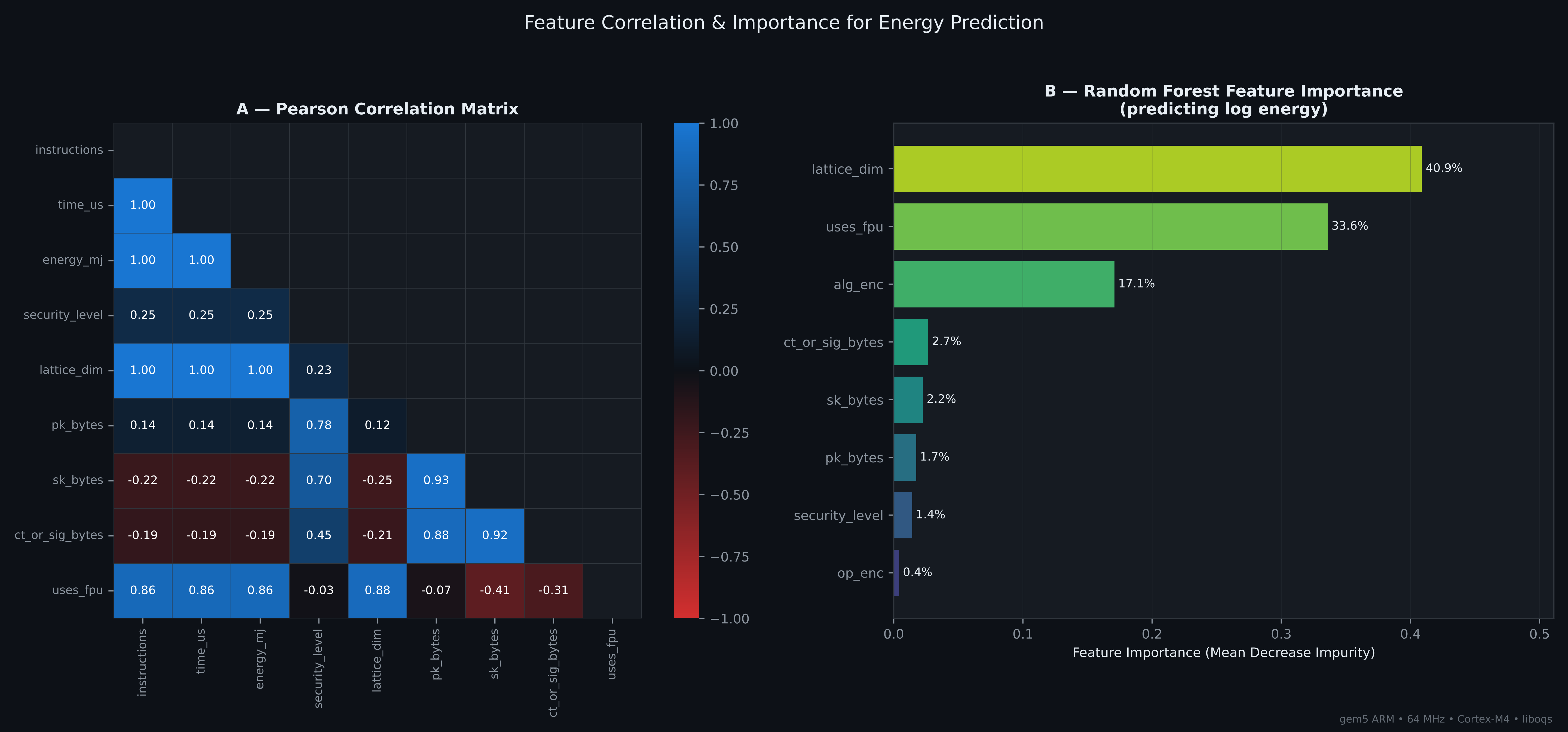


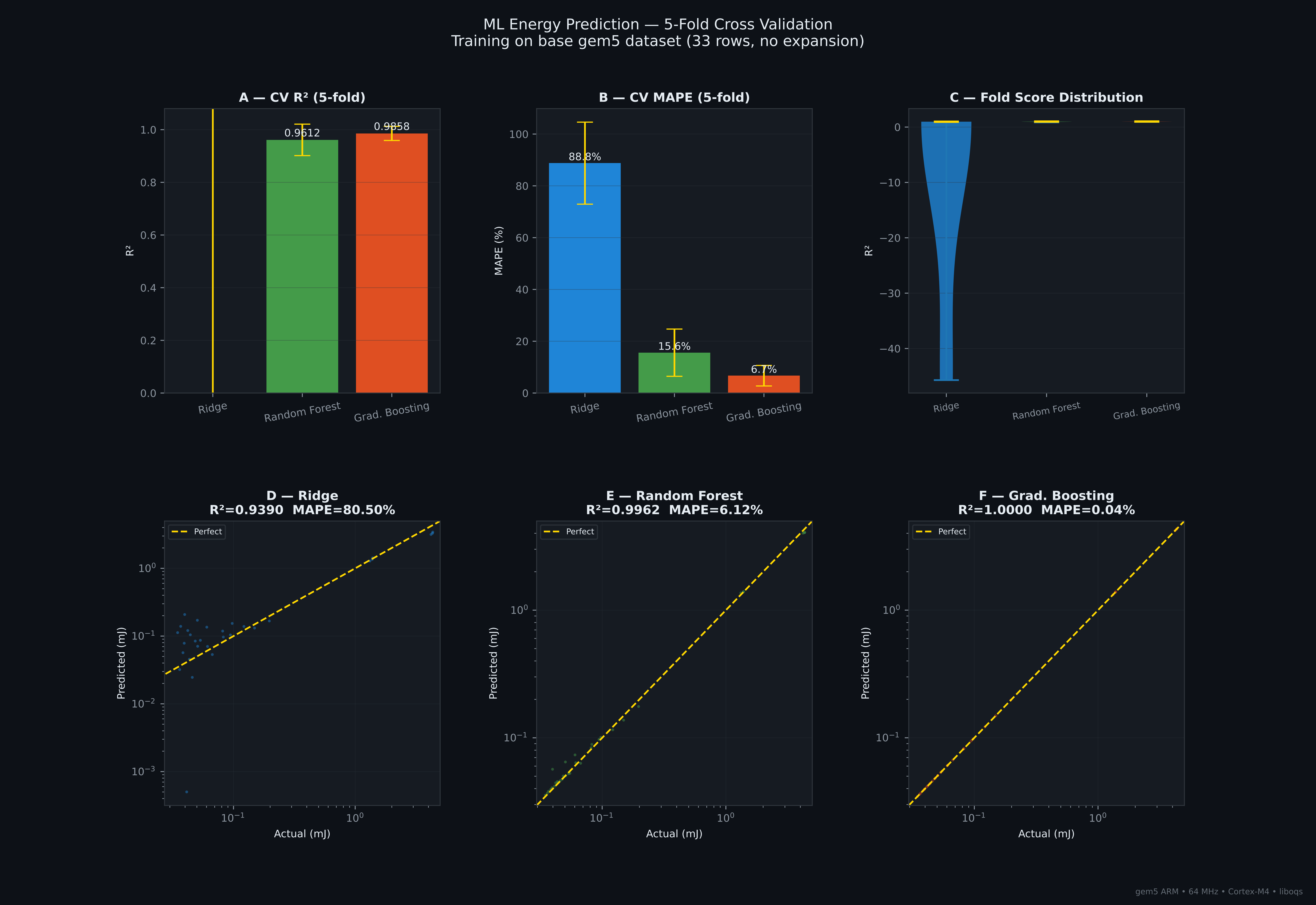
Energy Breakdown — KEM vs SIG Families
A — Total Energy by Family & Op
KEYGEN
ENCAP
DECAP
SIGN
VERIFY
Total Energy (mJ)
10^1
10^0
Falcon
Kyber
ML-DSA
ML-KEM
B — KEM vs SIG Distribution
log₁₀(Energy mJ)
0.5
0.0
−0.5
−1.0
−1.5
KEM
SIG
C — Energy Efficiency
(lower = better per security bit)
Falcon-512
Falcon-1024
ML-KEM-512
Kyber512
ML-DSA-44
ML-DSA-65
ML-DSA-87
ML-KEM-768
Kyber768
ML-KEM-1024
Kyber1024
KEM
SIG
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Total mJ / NIST Security Level
gem5 ARM • 64 MHz • Cortex-M4 • liboqs


Energy Heatmaps (mJ) — gem5 ARM Measurements
KEM Algorithms
Kyber1024
Kyber512
Kyber768
ML-KEM-1024
ML-KEM-512
ML-KEM-768
DECAP
ENCAP
KEYGEN
Operation
0.0603
0.0504
0.0397
0.0434
0.0393
0.0347
0.0507
0.0442
0.0368
0.0670
0.0534
0.0421
0.0475
0.0412
0.0361
0.0548
0.0458
0.0384
mJ
0.065
0.060
0.055
0.050
0.045
0.040
0.035
SIG Algorithms
Falcon-1024
Falcon-512
ML-DSA-44
ML-DSA-65
ML-DSA-87
KEYGEN
SIGN
VERIFY
Operation
4.1993
4.3167
4.3408
1.3211
1.3791
1.3915
0.0485
0.0815
0.0976
0.0612
0.0944
0.1218
0.0825
0.1490
0.1970
mJ
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
gem5 ARM • 64 MHz • Cortex-M4 • liboqs



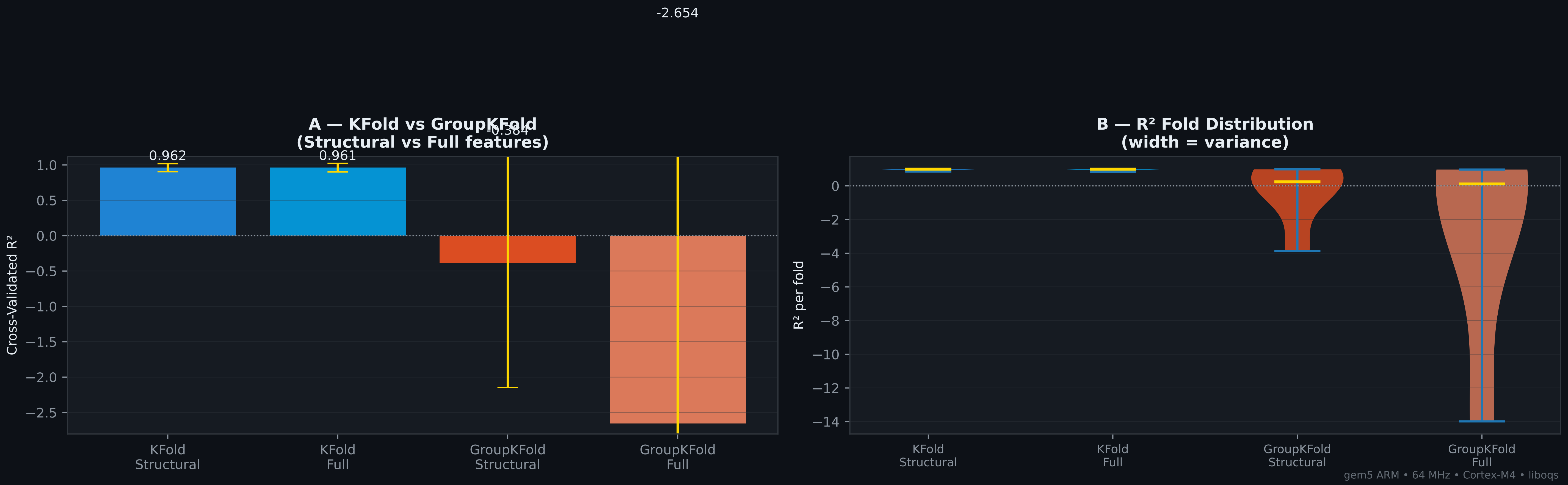
IoT Feasibility Analysis — ARM Cortex-M4 @ 64 MHz
A — Execution Time (μs)
algorithm
operation
DECAP
ENCAP
KEYGEN
SIGN
VERIFY
Falcon-1024 0 0 63625 65404 65770
Falcon-512 0 0 20017 20895 21084
Kyber1024 914 764 601 0 0
Kyber512 658 596 526 0 0
Kyber768 769 669 558 0 0
ML-DSA-44 0 0 735 1234 1479
ML-DSA-65 0 0 927 1430 1846
ML-DSA-87 0 0 1250 2258 2984
ML-KEM-1024 1015 809 638 0 0
ML-KEM-512 720 625 547 0 0
ML-KEM-768 831 695 582 0 0
μs
0
10000
20000
30000
40000
50000
60000
B — IoT-Feasible Ops per Algorithm
All feasible
Partial
None
ML-KEM-768 100%
ML-KEM-512 100%
ML-KEM-1024 100%
ML-DSA-87 100%
ML-DSA-65 100%
ML-DSA-44 100%
Kyber768 100%
Kyber512 100%
Kyber1024 100%
Falcon-512 67%
Falcon-1024 67%
IoT-Feasible Operations (out of 3)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
C — IoT-Feasible Operations Only
Falcon
Kyber
ML-DSA
ML-KEM
Falcon-1024
Falcon-512
ML-DSA-87
VERIFY
ML-DSA-87
SIGN
Energy (mJ)
Time (μs)
10^0
10^{-1}
10^3
10^4
D — IoT Feasibility by Security Level
Feasible
Infeasible
Number of Operations
NIST Security Level
0
2
4
6
8
10
12
1
2
3
5
gem5 ARM • 64 MHz • Cortex-M4 • liboqs


Feature Correlation & Importance for Energy Prediction
A — Pearson Correlation Matrix
instructions
time_us
energy_mj
security_level
lattice_dim
pk_bytes
sk_bytes
ct_or_sig_bytes
uses_fpu
1.00
1.00 1.00
0.25 0.25 0.25
1.00 1.00 1.00 0.23
0.14 0.14 0.14 0.78 0.12
-0.22 -0.22 -0.22 0.70 -0.25 0.93
-0.19 -0.19 -0.19 0.45 -0.21 0.88 0.92
0.86 0.86 0.86 -0.03 0.88 -0.07 -0.41 -0.31
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
B — Random Forest Feature Importance
(predicting log energy)
lattice_dim 40.9%
uses_fpu 33.6%
alg_enc 17.1%
ct_or_sig_bytes 2.7%
sk_bytes 2.2%
pk_bytes 1.7%
security_level 1.4%
op_enc 0.4%
0.0
0.1
0.2
0.3
0.4
0.5
Feature Importance (Mean Decrease Impurity)
gem5 ARM • 64 MHz • Cortex-M4 • liboqs



ML Energy Prediction — 5-Fold Cross Validation
Training on base gem5 dataset (33 rows, no expansion)
A — CV R² (5-fold)
R²
0.9612
0.9858
Ridge
Random Forest
Grad. Boosting
B — CV MAPE (5-fold)
MAPE (%)
88.8%
15.6%
6.7%
C — Fold Score Distribution
D — Ridge
R²=0.9390 MAPE=80.50%
E — Random Forest
R²=0.9962 MAPE=6.12%
F — Grad. Boosting
R²=1.0000 MAPE=0.04%
Perfect
Predicted (mJ)
Actual (mJ)
gem5 ARM • 64 MHz • Cortex-M4 • liboqs

Experiment B — Cross-Algorithm Generalisation (GroupKFold)
Tests whether model predicts unseen algorithms from structure alone



Summary — gem5 ARM Results & ML Model Performance

A — All gem5 Measurements (33 rows)

Algorithm	Op	Instructions	Time (µs)	Energy (mJ)	IoT
Kyber1024	DECAP	1,666,035	914.1	0.0603	✓
Kyber1024	ENCAP	1,411,418	764.4	0.0504	✓
Kyber1024	KEYGEN	1,131,129	601.4	0.0397	✓
Kyber512	DECAP	1,230,556	657.9	0.0434	✓
Kyber512	ENCAP	1,125,732	596.2	0.0393	✓
Kyber512	KEYGEN	1,004,680	526.2	0.0347	✓
Kyber768	DECAP	1,420,996	768.9	0.0507	✓
Kyber768	ENCAP	1,251,368	669.3	0.0442	✓
Kyber768	KEYGEN	1,058,582	557.9	0.0368	✓
ML-KEM-1024	DECAP	1,888,189	1014.7	0.0670	✓
ML-KEM-1024	ENCAP	1,517,961	809.4	0.0534	✓
ML-KEM-1024	KEYGEN	1,209,103	637.6	0.0421	✓
ML-KEM-512	DECAP	1,358,552	719.7	0.0475	✓
ML-KEM-512	ENCAP	1,187,481	624.9	0.0412	✓
ML-KEM-512	KEYGEN	1,046,730	546.8	0.0361	✓
ML-KEM-768	DECAP	1,556,727	830.8	0.0548	✓
ML-KEM-768	ENCAP	1,311,387	694.5	0.0458	✓
ML-KEM-768	KEYGEN	1,109,614	582.2	0.0384	✓
Falcon-1024	KEYGEN	124,847,496	63625.1	4.1993	✗
Falcon-1024	SIGN	127,775,399	65404.3	4.3167	✓
Falcon-1024	VERIFY	128,457,057	65770.1	4.3408	✓
Falcon-512	KEYGEN	39,296,848	20017.4	1.3211	✗
Falcon-512	SIGN	40,755,364	20895.4	1.3791	✓
Falcon-512	VERIFY	41,104,846	21083.7	1.3915	✓
ML-DSA-44	KEYGEN	1,383,960	734.6	0.0485	✓
ML-DSA-44	SIGN	2,276,270	1234.3	0.0815	✓
ML-DSA-44	VERIFY	2,717,016	1479.0	0.0976	✓
ML-DSA-65	KEYGEN	1,729,955	926.5	0.0612	✓
ML-DSA-65	SIGN	2,633,678	1429.6	0.0944	✓
ML-DSA-65	VERIFY	3,385,397	1846.1	0.1218	✓
ML-DSA-87	KEYGEN	2,311,900	1249.5	0.0825	✓
ML-DSA-87	SIGN	4,125,784	2257.7	0.1490	✓
ML-DSA-87	VERIFY	5,437,585	2984.4	0.1970	✓

B — ML Results & IoT Feasibility

Model	CV R^2 Mean	± Std	CV MAPE	Note
Ridge	-8.3709	± 18.6673	88.76%	
Random Forest	0.9612	± 0.0598	15.58%	
Grad. Boosting	0.9858	± 0.0268	6.69%	★ Best

IoT Feasibility Summary

```
✓  Kyber512      3/3 ops feasible
✓  Kyber768      3/3 ops feasible
✓  Kyber1024     3/3 ops feasible
✓  ML-KEM-512    3/3 ops feasible
✓  ML-KEM-768    3/3 ops feasible
✓  ML-KEM-1024   3/3 ops feasible
✓  ML-DSA-44     3/3 ops feasible
✓  ML-DSA-65     3/3 ops feasible
✓  ML-DSA-87     3/3 ops feasible
~  Falcon-512    2/3 ops feasible
~  Falcon-1024   2/3 ops feasible
```